

Current MoE Reasoning Experiments

1 Scope

This report tracks the three active workflows: `gepaLLMAsJudge`, `probeTest`, and `correlation_pipeline`. The first two were called Experiments 0a and 0b in the previous combined manuscript. Their full protocols, intermediate results, tables, limitations, and external-validation results are retained here under their current names.

The previous combined manuscript for numbered Experiments 1–5 is preserved at `old/report/legacy_full_report.tex` together with its compiled PDF. Those archived results remain available for provenance but are not mixed into the current experimental plan.

2 Experimental Setup

Current experimental scope. The active repository contains three complementary Qwen workflows. `gepaLLMAsJudge` is an annotation-capability study that optimizes a Qwen3.6-27B prompt for sentence-level Schoenfeld episode classification. It does not provide evidence about Qwen router dynamics. `probeTest` is a representation-probe replication that teacher-forces the 38 released DeepSeek-R1 Schoenfeld traces through the official GPTQ 4-bit Qwen/Qwen3.5-35B-A3B-GPTQ-Int4 checkpoint and tests whether the next sentence’s label is linearly decodable at its causal pre-sentence boundary. The separate `correlation_pipeline` generates benchmark solutions with Qwen3.5-35B-A3B, replays them through the matching checkpoint, and relates router and hidden-state summaries to correctness and reasoning-event labels.

GEPA judge design. The GEPA workflow starts from 3,125 annotated units in 38 released DeepSeek-R1 SAT responses. Complete responses are assigned to partitions before their units are flattened, preventing neighboring units from crossing splits. A local `llama.cpp` server supplies Qwen3.6-27B predictions. Each request contains the SAT problem and the previous, current, and next response units. The context-aware seed instruction contains 21 audited synthetic contrastive demonstrations. The implementation supports inverse-frequency weighted exact match and an informative same-model LLM-as-a-judge reward. Prompt-selection rules are evaluated with nested response-grouped cross-validation; the later final-fit run uses a response-grouped 26/6/6 train/validation/test split.

Schoenfeld boundary-probe design. The probe workflow excludes the 38 synthetic `</think>` closing units, which are all labelled Monitor, leaving the paper-matched 3,087 units. Qwen receives each SAT instruction as the user message and the released response as a teacher-forced assistant continuation. For a target unit, the feature is the hidden state immediately before its first token, so the probe sees the problem and preceding response but cannot read the target sentence. Hidden-state index 0 is the embedding output and indices 1–40 are the successive language-model layers.

At each index, seven separate target-versus-rest logistic regressions use raw 2,048-dimensional activations, balanced class weights, L2 regularization with $C = 1$, and no standardization or PCA. The paper-matched protocol uses a target-stratified 80/20 sentence split with seed 42. This is a replication choice rather than a strong

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generalization design: all 38 response IDs occur in both partitions. The reported layer for each target maximizes accuracy on that same test split. For qualitative transfer inspection, the selected probes are applied to 20 examples each from GSM8K, MATH, PRM800K, and ProcessBench. The seven binary scores are independently trained and uncalibrated; their argmax is an inspection label, not a multiclass probability or new gold annotation. Only boundary vectors are stored; Qwen does not regenerate the released or benchmark solutions.

Correlation-pipeline design. The pipeline separates llama.cpp generation from a teacher-forced Hugging Face forward pass. It stores router logits, selected expert identifiers, and optional hidden states, then computes trace-level and repeated-attempt correlations with correctness and lexical reasoning-event labels. The completed run covers 500 MATH-500 problems with one sample per problem. Generation uses the UD-Q4_K_XL GGUF with embedded native MTP speculative weights; replay uses the matching BF16 checkpoint with runtime 4-bit quantization, so the stages use different 4-bit formats. The remaining benchmarks and repeated-attempt evaluations are pending.

Table model and metric conventions. Every table caption identifies the evaluated model or models and the metrics shown. For classification, accuracy is the fraction of exact label matches, balanced accuracy is the unweighted mean of class recalls, precision and recall have their standard true-positive definitions, macro-F1 is the unweighted mean of per-class F1, and AUROC measures score ranking across the positive and negative classes. Cohen’s κ is chance-corrected nominal agreement, Kendall’s τ_b is rank agreement under the fixed episode-label ordering, and the GEPA composite averages their equal-weight $[0, 1]$ rescalings. Routing entropy is Shannon entropy of the softmax router distribution; switch rate is the fraction of top- k experts replaced from the preceding token; top- k overlap is adjacent-token Jaccard overlap; and router margin is the top-1 minus top-2 router probability. Unless a caption states otherwise, N and support are observation counts, r is a Pearson-family correlation coefficient, and confidence intervals are percentile bootstrap intervals.

Artifact provenance. The active packages are located at src/moe_exp/gepaLLMAsJudge, src/moe_exp/probeTest, and src/moe_exp/correlation_pipeline. GEPA runs through run_gepaLLMAsJudge.sh and writes to results/gepaLLMAsJudge; boundary-probe outputs remain under results/probeTest. The correlation workflow is launched by its local run_all.sh and uses its documented output directory.

The Qwen judge artifacts formerly called Experiment 0a and the Qwen3.5 boundary-probe artifacts formerly called Experiment 0b are complete. The MATH-500 correlation generation, replay, and analysis artifacts are complete for the single-attempt protocol. Historical OLMoE and numbered-experiment outputs are archived and are not pooled with the estimates in this report.

3 gepaLLMAsJudge

The gepaLLMAsJudge workflow evaluates whether prompt optimization improves a local judge for the seven adapted Schoenfeld episode labels: Read, Analyze, Plan, Implement, Explore, Verify, and Monitor. The model was the Q4_K_XL quantization of Qwen3.6-27B served by llama.cpp on one RTX 3090. All splits preserve complete responses, preventing neighboring units from crossing partitions.

Preliminary single-split runs. With seed 42, the 38 responses were divided into 26 training, 6 validation, and 6 test documents (2,382/407/336 units). These first runs classified the current unit without neighboring context and optimized per-unit exact match.

The matched few-shot run used the same seed, document split, model, and GEPA budget, adding seven gold training examples to the instruction. It performs better than the optimized base condition on the held-out test set: accuracy rises from 66.07% to 69.94% (+3.87 percentage points), Cohen’s κ from 0.5710 to 0.6178, Kendall’s τ_b from 0.6066 to 0.6667, and composite agreement from 0.7944 to 0.8211. GEPA itself does not improve the few-shot

Table 1. Preliminary gepaLLMAsJudge agreement. The composite rescales Cohen’s κ and Kendall’s τ_b from $[-1, 1]$ to $[0, 1]$ and averages them. Coverage was 100% in every row. *Model(s)*: Qwen3.6-27B Q4_K_XL, used as the episode-label judge. *Metrics*: exact-match accuracy, Cohen’s κ , Kendall’s τ_b , their rescaled composite, and valid-output coverage.

Evaluation	Accuracy	κ	τ_b	Composite
Seed validation	0.6388	0.5581	0.4810	0.7598
Optimized validation	0.6855	0.6074	0.5847	0.7980
Optimized test	0.6607	0.5710	0.6066	0.7944
Few-shot validation	0.6806	0.6078	0.5830	0.7977
Few-shot test	0.6994	0.6178	0.6667	0.8211

Table 2. Aggregate selected-prompt outer-fold results over 2,789 units. “Balanced light” is the earlier 0.10 recall-gate run; the macro-F1 runs use a 0.05 gate. The six excluded responses were not evaluated. *Model(s)*: Qwen3.6-27B Q4_K_XL, used as the episode-label judge. *Metrics*: exact-match accuracy, balanced accuracy, macro-F1, and the rescaled κ/τ_b composite.

Selection / budget	Accuracy	Balanced acc.	Macro-F1	Composite
Balanced / light	0.6701	0.6753	0.6480	0.8031
Macro-F1 / light	0.6734	0.6796	0.6505	0.8063
Macro-F1 / heavy	0.6669	0.6737	0.6437	0.8030

validation result: the seed and selected optimized prompts both score 68.06% accuracy. The current comparison therefore favors few-shot prompting for held-out generalization, whereas the measurable GEPA optimization gain is confined to the base-prompt condition.

Context-aware nested cross-validation. The revised protocol excludes the same six responses from model selection and runs five outer folds over the remaining 32 responses. Within each fold, five responses form the inner validation set and all others outside the outer fold form the GEPA training set. Each request includes the problem statement, previous unit, current unit, and next unit. The seed instruction includes 21 audited synthetic contrastive demonstrations. GEPA receives inverse-frequency weighted exact match, whose split mean is balanced accuracy; the safety selector compares either balanced accuracy or macro-F1 and rejects any candidate whose class recall falls by more than the configured threshold.

Informative same-model reward. The current implementation can replace the static GEPA reward with an LLM-as-a-judge call to the same Qwen3.6-27B model. For each candidate label, the judge returns a short free-text explanation plus four diagnostic scores from 1 to 5: gold agreement, functional fit, contextual coherence, and boundary precision. Their normalized weighted mean is used only as GEPA’s optimization signal, with respective weights 0.40, 0.25, 0.15, and 0.20. Gold-label accuracy, balanced accuracy, macro-F1, and per-class recall remain the independent selection and reporting criteria. The judge runs deterministically with model reasoning disabled. Every judge call is persisted as JSONL with its component scores, normalized score, and free-text explanation, permitting a direct audit of feedback quality. Tables 2 and 3 remain static-reward cross-validation baselines; the completed LLM-judge final fit is reported separately below because its reused validation set is a prompt-development set rather than an outer-fold estimate.

Macro-F1 selection with the light budget is the strongest cross-validated configuration, but its advantage over balanced-accuracy selection is small: +0.32 accuracy points, +0.43 balanced-accuracy points, and +0.24 macro-F1 points. At the response level it wins on 15 documents, loses on 13, and ties on 4, so the observed gain is not stable

Table 3. Per-class outer-fold performance for the best current configuration, macro-F1 selection with the light GEPA budget. *Model(s)*: Qwen3.6-27B Q4_K_XL, used as the episode-label judge. *Metrics*: class support, precision, recall, and F1.

Class	Support	Precision	Recall	F1
Read	318	0.798	0.770	0.784
Analyze	685	0.757	0.632	0.689
Plan	205	0.524	0.590	0.555
Implement	650	0.880	0.722	0.793
Explore	193	0.679	0.492	0.571
Verify	538	0.669	0.606	0.636
Monitor	200	0.364	0.945	0.526

across documents. The light run selects a GEPA-optimized prompt in three of five folds; the recall gate retains the seed prompt in the other two because Plan or Monitor recall falls too far.

Increasing the budget does not help. The heavy run selects an optimized prompt in only two folds and is 0.65 accuracy points and 0.68 macro-F1 points below the light run. Paired predictions show 38 units correct only under light versus 20 correct only under heavy. This difference is concentrated in fold 5, where light discovers and selects a strong candidate whereas heavy retains the seed after its candidate violates the Analyze recall gate. Independent heavy search is therefore not a superset of light search and can discard a better prompt found by a smaller run.

The revised objective improves overall balance only marginally and does not resolve Monitor overprediction. The model emits Monitor 519 times for 200 gold Monitor units; the leading errors are Verify→Monitor (127), Plan→Monitor (65), and Explore→Monitor (42). The annotation audit flags 180/3,125 units (5.76%), including 145 multiline units and 49 structural/control markers such as </think> and “Final Answer.” These artifacts plausibly inflate Monitor recall while depressing precision and should be reviewed before further optimization.

All runs retain 100% final-output coverage and complete without OOM failures, but the light and heavy jobs log 31 and 28 truncated LM responses during optimization. Moreover, the outer folds currently evaluate only the safety-selected prompt, not both seed and optimized prompts, so they do not directly estimate GEPA’s paired outer-fold gain. Finally, the six responses excluded from current CV were inspected in the preliminary experiments; they are locked from the current code path but no longer constitute a pristine research holdout. The judge should therefore not replace validated annotation until structural artifacts are adjudicated, decoding is constrained, paired outer evaluation is added, and a genuinely unseen external test set is designated.

Completed LLM-judge final fit. The August 11, 2026 run uses seed 42, the 21-example context-aware few-shot prompt, and the same-model informative reward on the original response-grouped 26/6/6 split (2,382/407/336 units). Seed and optimized prompts both return a valid label for all 407 validation units. Table 4 reports exact gold-label metrics rather than the LLM judge’s diagnostic score.

The optimized prompt fixes 49 seed errors and regresses on 14 previously correct units, a net gain of 35. All six validation responses improve, with response-level accuracy gains from 2.0 to 13.8 percentage points. Its primary effect is to curb the seed prompt’s overuse of Monitor: Monitor predictions fall from 74 to 43, including nine corrections from Monitor to true Plan and 15 from Monitor to true Verify. Recall rises most for Plan (+28.13 points), Explore (+23.53), Verify (+14.29), and Analyze (+8.20). Monitor recall falls slightly from 1.000 to 0.944, but precision rises from 0.243 to 0.395. The selected prompt still has weak Monitor precision and Analyze recall, and the remaining errors concentrate at the Analyze/Implement/Verify boundaries and Verify→Monitor. Implement

Table 4. Context-aware LLM-judge final-fit results on the 407-unit prompt-development validation set. Coverage is 100% in both rows. The locked test is not evaluated. *Model(s)*: Qwen3.6-27B Q4_K_XL, used both for episode-label prediction and the same-model GEPA reward. *Metrics*: exact-match accuracy, balanced accuracy, macro-F1, Cohen’s κ , Kendall’s τ_b , absolute change, and valid-output coverage.

Prompt	Accuracy	Balanced acc.	Macro-F1	κ	τ_b
21-example seed	0.6683	0.6765	0.6132	0.5988	0.5904
GEPA optimized	0.7543	0.7769	0.7270	0.6968	0.6832
Absolute change	+0.0860	+0.1003	+0.1138	+0.0980	+0.0929

Table 5. Per-class recall and optimized F1 for the completed LLM-judge final fit. *Model(s)*: Qwen3.6-27B Q4_K_XL, used both for episode-label prediction and the same-model GEPA reward. *Metrics*: class support, seed and optimized recall, and optimized per-class F1.

Class	Support	Seed recall	Opt. recall	Opt. F1
Read	45	0.778	0.756	0.791
Analyze	122	0.598	0.680	0.744
Plan	32	0.531	0.813	0.722
Implement	103	0.786	0.825	0.825
Explore	17	0.471	0.706	0.686
Verify	70	0.571	0.714	0.763
Monitor	18	1.000	0.944	0.557

is the only class whose F1 decreases, from 0.839 to 0.825, because its additional predictions include more false positives.

The internal same-model reward rises from 0.7264 for the seed to 0.8196 for the selected candidate. The optimization trace contains 145 candidates, 144 full validation evaluations, and 60,767 persisted judge calls, of which 148 are judge-output errors. A text audit finds 37 validation units for which the judge explicitly disputes the corpus gold label in at least one evaluation, showing that the reward itself detects annotation-sensitive boundaries rather than consistently treating gold as semantically authoritative. The structural audit flags 29 validation units; their accuracy remains 72.4% under both prompts, so the gain comes entirely from otherwise unflagged units.

GEPA expands the instruction from 21 to 40 demonstrations and from 7,648 to 32,717 characters. The extra rules and examples target observed corpus edge cases, raising inference cost and the risk of learning annotation conventions rather than a portable episode taxonomy. More importantly, GEPA evaluates 145 candidates against the same six-response validation set used for final selection. The nominal 407 units are also correlated within responses, and only one optimization seed is available. The result files explicitly record `locked_test_evaluated: false` and `test: null`. The observed gain is therefore strong evidence of successful prompt optimization on the development distribution, but not a held-out generalization estimate. The prompt should be frozen and compressed or ablated before a one-time test on genuinely unseen responses.

4 probeTest

The probeTest workflow tests a different annotation mechanism from the generative judge in `gepaLLMAsJudge`. It asks whether Qwen’s representation of the prefix immediately before a sentence contains linearly accessible

Table 6. Accuracy-selected Schoenfeld boundary probes on the sentence-level test split. Prevalence is over all 3,087 units. Δ Maj. subtracts the always-negative accuracy. Balanced accuracy is derived from the saved test confusion matrix. Layer selection and evaluation use the same test split and are therefore selection-optimistic. *Model(s)*: Qwen/Qwen3.5-35B-A3B-GPTQ-Int4 as a frozen hidden-state encoder, with one-versus-rest logistic probes. *Metrics*: target prevalence, selected hidden-state layer, accuracy, accuracy minus the majority baseline, balanced accuracy, F1, and AUROC.

Target	Prevalence	Layer	Accuracy	Δ Maj.	Balanced acc.	F1	AUROC
Read	12.21%	34	0.9239	+0.0461	0.8476	0.7044	0.9338
Analyze	25.62%	36	0.8107	+0.0669	0.7669	0.6465	0.8348
Plan	7.29%	39	0.9175	-0.0096	0.7405	0.4848	0.8551
Implement	21.87%	38	0.8964	+0.1151	0.8617	0.7714	0.9305
Explore	6.74%	40	0.9223	-0.0103	0.7486	0.4894	0.8884
Verify	20.38%	27	0.8754	+0.0792	0.8509	0.7260	0.9122
Monitor	5.90%	35	0.9304	-0.0106	0.7546	0.4819	0.9031

information about the upcoming Schoenfeld episode. It therefore measures prospective prefix-state decodability, not semantic classification after reading the sentence and not the latent dynamics of a Qwen-generated trace.

Within-corpus results. Table 6 reports the accuracy-selected layer for each one-versus-rest target. AUROC is 0.835–0.934 and balanced accuracy is 0.740–0.862, demonstrating substantial within-split ranking signal. Implement is strongest at the natural threshold (F1 0.771), followed by Verify (0.726) and Read (0.704). Analyze is intermediate. Plan, Explore, and Monitor have F1 below 0.50. Their nominal accuracies are also 0.96–1.06 percentage points below an always-negative majority classifier, so accuracy alone gives a misleading impression for these rare targets.

The signal emerges rapidly and is not maximal at the model output. Mean one-versus-rest AUROC across targets rises from 0.624 at the embedding output to 0.873 at layer 10 and 0.892 at layer 20, peaks at 0.899 at layer 30, and falls to 0.867 at layer 40. Mean F1 peaks at layer 28. Accuracy-based target selection sometimes obscures this profile: for Plan, the selected layer 39 has AUROC 0.855 whereas layer 21 reaches 0.922. AUROC or balanced accuracy, chosen inside a validation fold, better matches the scientific question of decodability.

The fits are numerically stable—none of the 287 regressions reports a convergence warning—but statistical overfitting remains visible. At the selected layers, train–test accuracy gaps are 16.7 points for Analyze, 10.1 for Implement, approximately 7.8 for Plan and Explore, and 4.5–5.5 for the other targets. Explore reaches 100% training accuracy. With 2,048 features, 2,469 training units, strong within-response dependence, and only 38 distinct responses, the effective independent sample size is much closer to the number of responses than to the number of sentences. Test-based layer selection adds further optimism.

External benchmark inspection. All 80 requested examples complete without omitted units, yielding 581 sentence boundaries. Table 7 shows that the seven-way inspection behavior collapses mainly to Implement, Read, and Monitor. Implement alone receives 49.2% of argmax assignments, whereas Explore is never selected. The transfer distribution also differs sharply from training: Analyze falls from 25.6% of gold units to 7.7% of predictions and Verify from 20.4% to 3.3%.

The independent binary scores do not form a clean multiclass decision rule. Seventy-four units (12.7%) have no score above 0.5 but are still assigned an argmax label, while 180 (31.0%) activate more than one probe and 99 (17.0%) have a top-two margin below 0.1. A conservative display rule requiring a top score of at least 0.8 and a

Table 7. Argmax inspection-label counts on external reasoning units. These are raw prediction frequencies, not accuracies. Table 8 compares them with the subsequent provisional pragmatic review. *Model(s)*: Qwen/Qwen3.5-35B-A3B-GPTQ-Int4 as a frozen encoder, with the seven selected one-versus-rest logistic probes. *Metrics*: unit counts and seven-way argmax prediction counts.

Dataset	Units	Read	Analyze	Plan	Implement	Explore	Verify	Monitor
GSM8K	96	0	2	3	53	0	0	38
MATH	88	23	7	1	45	0	1	11
PRM800K	139	34	18	0	71	0	1	15
ProcessBench	258	69	18	12	117	0	17	25
Total	581	126	45	16	286	0	19	89

Table 8. Agreement with the provisional pragmatic review. Primary accuracy requires the preferred label; compatible accuracy accepts either label on a multi-label unit. Macro-F1 and macro recall treat each accepted label as a positive reference membership. *Model(s)*: Qwen/Qwen3.5-35B-A3B-GPTQ-Int4 as a frozen encoder, with the seven selected one-versus-rest logistic probes. *Metrics*: unit count, primary accuracy, compatible accuracy, multi-label count, accepted-membership macro-F1, and macro recall.

Dataset	Units	Primary acc.	Compatible acc.	Multi-label	Macro-F1	Macro recall
GSM8K	96	49.0%	62.5%	20	0.136	0.129
MATH	88	40.9%	53.4%	36	0.211	0.228
PRM800K	139	46.0%	49.6%	44	0.241	0.302
ProcessBench	258	56.2%	68.6%	87	0.396	0.350
Total	581	50.3%	60.8%	187	0.330	0.306

margin of at least 0.2 retains only 318/581 units (54.7%), mostly Implement, Read, and Monitor. This threshold is a useful abstention heuristic, not a calibrated guarantee of correctness.

Provisional pragmatic adjudication. To make the qualitative inspection measurable, every displayed unit was subsequently reviewed using the problem, the current unit, its neighbors, and the adapted Schoenfeld precedence rule. The annotation is a hybrid rule-assisted Codex pragmatic review, not human consensus or authoritative benchmark gold. It records one primary label for conventional single-label statistics and, where two functions are genuinely defensible, an accepted set containing both labels. This occurs for 187/581 units (32.2%), mostly at boundaries where a unit both announces and executes an operation or where a structural marker also carries a direct result. Of the annotations, 394 have high, 150 medium, and 37 low confidence; the low-confidence cases remain priority items for human review.

The probe agrees with the primary pragmatic choice on 292/581 units (50.3%) and with at least one accepted label on 353/581 (60.8%). Thus recognizing genuine taxonomy overlap recovers 61 additional compatible decisions, but it does not make the reference authoritative. ProcessBench has the strongest compatible agreement (68.6%), followed by GSM8K (62.5%); MATH has the lowest strict agreement (40.9%). Table 8 reports the complete dataset-level results.

The class-level comparison in Table 9 sharpens the interpretation. Implement is the only operationally strong external label (precision 0.804, recall 0.667, F1 0.729). Read is moderately useful (F1 0.524). Analyze and Plan are severely under-produced, while Verify and Monitor have low recall. Explore has 11 accepted reference

Table 9. Per-label comparison with accepted pragmatic memberships. Support counts every unit for which the label is accepted, including secondary labels on ambiguous units. *Model(s)*: Qwen/Qwen3.5-35B-A3B-GPTQ-Int4 as a frozen encoder, with the seven selected one-versus-rest logistic probes. *Metrics*: accepted-label support, argmax prediction count, precision, recall, and F1.

Label	Accepted support	Probe predictions	Precision	Recall	F1
Read	84	126	0.437	0.655	0.524
Analyze	84	45	0.356	0.190	0.248
Plan	90	16	0.375	0.067	0.113
Implement	345	286	0.804	0.667	0.729
Explore	11	0	0.000	0.000	0.000
Verify	51	19	0.632	0.235	0.343
Monitor	103	89	0.382	0.330	0.354

memberships but is never predicted, hence zero recall. Because multi-label units contribute to two reference classes, accepted supports overlap and do not sum to 581.

Pragmatically, Implement is the most coherent transferred label: it marks long calculation, substitution, and algebra runs. Read often marks givens or restatements, particularly near the start of MATH and PRM800K solutions. Monitor behaves primarily as a solution-boundary or opening-state detector: all 20 GSM8K first units and 18/20 ProcessBench first units receive Monitor, whereas 19/20 MATH and 17/20 PRM800K first units receive Read. Because the target sentence is unseen at this point, this source dependence must arise from the problem, prefix position, and formatting rather than the displayed unit’s semantics. Analyze and Plan appear plausibly on some explanatory or phase-change boundaries but have low coverage. Explore is unusable for external tagging in the present model.

Verify has a narrower but potentially useful role as a review or reconsideration flag. All 17 ProcessBench Verify predictions occur at or after the benchmark’s annotated first error, including six within one trace that repeatedly reinterprets the question before ending with the wrong sign. This is not an error detector: errors generally occur early, Verify may merely mark late position or reflective language, and no PRM800K post-error unit is labelled Verify. It can prioritize passages for inspection but cannot judge correctness.

The qualitative stage also exposes an input problem independent of the classifier. Thirteen units are split immediately after abbreviations such as “Mr.”, producing fragments including “Therefore, Mr.” and inconsistent scores. External use should first adopt abbreviation- and LaTeX-aware segmentation. With the current outputs, the defensible practical representation is a coarse three-phase view—setup (Read and opening Monitor), execution (Analyze, Plan, and Implement), and review (Verify)—plus an explicit unknown/ambiguous state. The seven raw argmax labels should not be used as automatic cognitive annotations.

5 Qwen Correlation Pipeline

The correlation pipeline completed all 500 MATH-500 problems with one sampled trace per problem. Qwen3.5-35B-A3B answered 387 correctly (77.4%). The lexical detectors marked 133 traces with backtracking, 97 with self-correction, and 70 with contradiction. The analysis contains 59 trace features, including prespecified summaries across the six router layers selected by the independent Schoenfeld probe run. Table 10 reports point-biserial correlations; intervals resample source problems. Because there is exactly one trace per problem, the 500 bootstrap clusters are also the 500 observations.

Correct traces are shorter and exhibit lower mean router entropy, smaller hidden-state norms and trajectory steps, less top-1 switching, greater top- k overlap, and larger router margins. The entropy association is the largest

Table 10. Prespecified Qwen MATH-500 trace-level correlations. Positive r means larger feature values in traces for which the binary target is present. Intervals are 95% problem-cluster bootstrap intervals. *Model(s)*: Qwen3.5-35B-A3B (UD-Q4_K_XL GGUF generation and matching unsloth/Qwen3.5-35B-A3B routing replay with runtime 4-bit quantization). *Metrics*: point-biserial r between each binary trace target and the listed continuous feature, with problem-cluster bootstrap 95% confidence intervals.

Target	Feature	r	95% CI
Correct answer	Mean router entropy	−0.555	[−0.614, −0.500]
Correct answer	Token count	−0.520	[−0.562, −0.480]
Correct answer	Mean hidden-state norm	−0.488	[−0.554, −0.419]
Correct answer	Mean router margin	0.301	[0.231, 0.376]
Correct answer	Mean hidden/router geometry	−0.238	[−0.307, −0.164]
Backtracking	Token count	0.453	[0.399, 0.501]
Backtracking	Mean router entropy	0.364	[0.268, 0.452]
Self-correction	Token count	0.382	[0.336, 0.429]
Self-correction	Mean router entropy	0.325	[0.224, 0.411]
Contradiction	Mean hidden-state norm	0.329	[0.259, 0.393]
Contradiction	Token count	0.247	[0.189, 0.302]

prespecified routing result. Exploratory layerwise screening yields similar or slightly larger magnitudes—entropy at layer 27 has $r = -0.574$, and hidden-state step distance at layer 39 has $r = -0.572$ —but these maxima do not have selection-adjusted intervals and should not be treated as confirmed layer localization.

The same direction recurs for the lexical reasoning-event targets: traces with backtracking or self-correction tend to be longer, higher-entropy, and lower-margin. However, token count is itself strongly associated with correctness and is the largest correlate of backtracking and self-correction. The present analysis reports marginal correlations and does not condition routing features on length, problem difficulty, or token content. It therefore cannot distinguish a routing-specific signal from longer or harder failed solutions having more opportunities for revision markers and different activation statistics. With one attempt per problem, problem-level pass-rate correlations and within-problem correct-minus-incorrect contrasts are empty; the repeated-attempt design is required for those stronger controls.

6 Analysis

6.1 Conclusions supported so far

- (1) The static-reward, context-aware GEPA study reaches 0.650 cross-validated macro-F1 with the light budget. The advantage over balanced-accuracy selection is small and the heavy search is worse. In the separate completed same-model LLM-judge final fit, development-set balanced accuracy rises from 0.677 to 0.777 and macro-F1 from 0.613 to 0.727, primarily through reduced Monitor overprediction. This establishes successful prompt optimization on the development distribution, not held-out generalization.
- (2) The boundary probes show substantial sentence-split decodability of upcoming Schoenfeld episodes, with selected-layer AUROC of 0.835–0.934 and the strongest thresholded performance for Implement, Verify, and Read. This establishes accessible prefix-state signal within familiar responses, not generalization to unseen reasoning traces. External application is best interpreted as coarse phase inspection: a provisional pragmatic review gives 50.3% strict and 60.8% compatible agreement, with 187/581 multi-label units.

Implement is strongest (F1 0.729), while the remaining labels are incomplete external tags. ProcessBench Verify predictions identify review-heavy post-error passages but do not identify correctness.

- (3) In the completed 500-problem Qwen MATH-500 correlation run, correctness is associated with lower mean router entropy ($r = -0.555$, 95% CI $[-0.614, -0.500]$) and larger mean router margin ($r = 0.301$, $[0.231, 0.376]$). Correct traces are also substantially shorter ($r = -0.520$), while backtracking and self-correction are most strongly associated with greater length. These are marginal trace-level associations; the single-attempt design cannot isolate routing from length, token content, or problem difficulty.

6.2 What has not been established

The GEPA final fit does not provide an unbiased test estimate. It selects among 145 candidates on the same six-response validation set, has only one optimization seed, and leaves the locked 336-unit test unevaluated. The result does not establish that the expanded 40-demonstration prompt transfers to new responses or that the same-model reward is an independent evaluator.

The boundary-probe run does not establish response-grouped generalization, calibrated seven-way episode classification, or the cognitive states of Qwen’s own generations. It uses Qwen to encode teacher-forced DeepSeek-R1 text, shares all 38 response IDs between train and test, and selects layers on the reported test set. Its external argmax outputs have no authoritative human or benchmark episode ground truth. Agreement with the provisional hybrid review is diagnostic and should not be called validated transfer accuracy.

The Qwen MATH-500 correlations do not establish an independent routing predictor or a causal mechanism. No length-conditioned model has been fit, only one attempt is available per problem, and the other planned benchmarks have not yet been analyzed. Layerwise maxima are exploratory comparisons without selection-adjusted uncertainty. Across all three workflows, correlation and linear decodability should not be read as causal evidence.

7 Discussion

The completed GEPA final fit demonstrates that same-model informative feedback can guide a large prompt toward the development labels: it fixes 49 seed errors while introducing 14 regressions and improves every validation response. Its optimization trace also shows the central risk. The prompt expands from 21 to 40 demonstrations and is selected after 145 candidate evaluations on those same six responses. The next defensible test is therefore to freeze and ablate or compress the prompt, quantify response-level uncertainty, and evaluate it once on genuinely unseen responses.

The episode probes add a complementary representation-level result: the prefix before a sentence often encodes the phase likely to follow, and the signal is strongest in middle-to-late layers rather than at the final model output. The current split cannot determine whether this reflects a portable episode representation, trace-specific transition regularities, position, or formatting. A response-grouped nested evaluation should select both layer and regularization on inner folds, evaluate once on unseen responses, repeat over seeds, and report response-bootstrap intervals. Standardized or norm-matched features and simple position, previous-unit, and formatting baselines are needed to separate hidden-state information from layer scale and trace structure.

For external use, the probe outputs should be treated as abstaining phase suggestions rather than annotations. Abbreviation- and LaTeX-aware segmentation should precede evaluation; independent one-versus-rest scores should either be calibrated on a common multiclass validation set or retain multiple labels. The pragmatic review confirms substantial ambiguity and only 60.8% compatible agreement. Independent human annotators should resolve the 37 low-confidence units and assess inter-annotator agreement before any transfer-accuracy claim. Cross-model recognition, as studied here, should also be separated from Qwen’s own generation by running the protocol on Qwen-generated traces.

The Qwen MATH-500 run reinforces the need for structural controls. Lower entropy and higher margin accompany correct answers, but incorrect and revision-bearing traces are also longer. The appropriate next test is the planned repeated-attempt analysis: compare correct and incorrect samples of the same problem, condition on trace length and token composition, and check whether the aggregate entropy and margin effects replicate across datasets.

8 Conclusion

The three active workflows support deliberately limited claims. GEPA improves the same-model-judge development result substantially but has not been tested on an untouched response set. The boundary probes reveal strong within-split linear access to upcoming episode labels, especially for Implement, Verify, and Read, but their sentence split leaks every response across train and test and their external outputs collapse toward a coarse setup, execution, and review taxonomy. The Qwen MATH-500 study adds a robust descriptive association—correct traces have lower router entropy and higher margin—but co-occurring length effects and the single-attempt design prevent an independent or causal routing claim.

The immediate evidential requirements are a frozen-prompt GEPA evaluation on unseen responses, response-grouped nested boundary probes with independently validated external labels, and repeated-attempt benchmark generation enabling within-problem routing contrasts.

9 Immediate Work

The agreed order of work is:

- (1) run one final GEPA alignment with the tutor-confirmed checkpoint and medium reasoning budget, then freeze the selected prompt;
- (2) audit tensor semantics, shapes, dtypes, and serialized bytes per trace, then retain only justified router and hidden-state locations;
- (3) prespecify the association measures, uncertainty, clustering, and multiplicity treatment using cited methodology;
- (4) generate repeated attempts for AIME 2024/2025 and an affordable MATH-500 pilot, then evaluate within-problem routing contrasts; and
- (5) expand to additional MoE families only after the pilot pipeline and disk budget are stable.

The detailed acceptance criteria and transcript ambiguities are maintained in `NEXT_STEPS.md` at the repository root.

A Reproducibility Checklist

Table 11. Living inventory for the three active workflows. *Model(s)*: Qwen3.6-27B Q4_K_XL, Qwen/Qwen3.5-35B-A3B-GPTQ-Int4, and Qwen3.5-35B-A3B correlation-run variants. *Metrics*: workflow status, sample counts, headline metrics, and next actions; this is an inventory rather than a common-metric evaluation.

Item	Status	Note / next action
gepaLLMAsJudge	Cross-validation and final-fit validation complete; test pending	Static-reward grouped cross-validation over 2,789 units gives 67.34% accuracy and 65.05% macro-F1 for light-budget GEPA; heavy is worse. The same-model LLM-judge final fit improves validation accuracy from 66.83% to 75.43% and macro-F1 from 0.613 to 0.727, but reuses six responses across 145 candidate evaluations and does not evaluate the 336-unit locked test. Freeze and ablate the 40-example prompt, report response-level uncertainty, and evaluate on genuinely unseen responses.
probeTest	Complete; grouped and human validation pending	Qwen3.5 GPTQ probes cover 3,087 units, seven targets, and 41 hidden-state indices; selected-layer AUROC is 0.835–0.934. A provisional hybrid review of 581 external units gives 50.3% primary and 60.8% compatible agreement, with 187 multi-label and 37 low-confidence units; Explore is never predicted. Grouped cross-validation, robust segmentation, multiclass calibration, and independent human adjudication remain pending.
correlation_pipeline	MATH-500 complete; broader study pending	Generation, replay, and analysis are complete for all 500 MATH-500 problems (one attempt each; 387 correct). Prespecified aggregate correlations use problem-cluster bootstrap intervals, but repeated-attempt problem-level and within-problem analyses are unavailable under this design. The other planned benchmarks remain pending. Generation uses the UD-Q4_K_XL GGUF; replay uses the matching checkpoint with runtime 4-bit quantization, so the two stages use different 4-bit formats.